

```

1  %-----
2  % Consumer Choice (Octave)
3  %-----
4
5  clear
6  clc
7
8  %Printing the outputs as a text file if ==1
9  output_file=0;
10 %LaTeX Interpreter if ==1 (requires LaTeX to be installed)
11 latex_interpreter=1;
12
13 %Problem Set-up
14 alpha=0.7;
15 %phi = @(x) -(x(1).^alpha).*(x(2).^(1-alpha));
16 function y = phi_fun(x, alpha)
17     y = -(x(1).^alpha).*(x(2).^(1-alpha)); %negative
18 endfunction %Octave: not "end" but "endfunction"
19 phi = @(x) phi_fun(x, alpha);
20
21 %Constraint: we consider the inequality constraint h(x) >=0
22 A=[1, 1]; %A=[p1, p2]
23 b=[1]; %b=m
24 h = @(x) b - A*x;
25 %function y = h_fun(x, A, b)
26 % y = b - A*x;
27 %endfunction %Octave: not "end" but "endfunction"
28 %h = @(x) h_fun(x, A, b);
29
30 lb=[0; 0];
31 ub=[1e10, 1e10];
32 x0=[0.1; 0.1];
33
34 %sqp
35 [x, obj, info, iter, nf, lambda] = sqp (x0, phi, [], h, lb,ub);
36
37
38 if output_file==1
39     dfile="consumer_choice_sqp.txt";
40     if exist(dfile, 'file')
41         delete(dfile);
42     end
43     diary(dfile)
44     diary on
45 end
46
47 %Output
48 fprintf('alpha=%.3f, p1=%.3f, p2=%.3f, m=%.3f\n',alpha,A(1),A(2),b);
49 fprintf('Optimal consumption: %.3f,%.3f\n',x(1),x(2));
50 fprintf('Utility: %.5f\n',-obj);
51 fprintf('Multiplier (budget const): %.5f\n',lambda(1));
52 fprintf('Multiplier (non-negativity const): %.5f, %.5f\n',lambda(2),lambda(3));
53
54 if output_file==1
55     diary off
56 end
57
58 %Plot

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59 x1_vec=linspace(0, b/A(1), 100);
60 x2_vec=linspace(0, b/A(2), 100);
61 [x1_mesh, x2_mesh] = meshgrid(x1_vec, x2_vec);
62 figure();
63 hold on;
64 grid on;
65 plot(x1_vec, (b-A(1)*x1_vec)/A(2), 'k', 'LineWidth', 1); %budget constraint
66 contour(x1_mesh, x2_mesh, (x1_mesh.^alpha).*(x2_mesh.^(1-alpha)), [(-obj), (-obj)], 'LineWidth',
67 1, 'LineColor', 'b');
68 contour(x1_mesh, x2_mesh, (x1_mesh.^alpha).*(x2_mesh.^(1-alpha)), [0.75*(-obj), 1.25*(-obj)],
69 'LineWidth', 1, 'LineColor', 'k', 'LineStyle', '--');
68 plot(x(1),x(2),'ro','LineWidth', 1);
69 xlim([0,b/A(1)]);
70 ylim([0,b/A(2)]);
71 if latex_interpreter==1
72     xlabel('$x_1$', 'Interpreter', 'latex', 'FontSize', 12);
73     ylabel('$x_2$', 'Interpreter', 'latex', 'FontSize', 12);
74     title('Consumer Choice', 'Interpreter', 'latex', 'FontSize', 14);
75 else
76     xlabel('x_1', 'FontSize', 12);
77     ylabel('x_2', 'FontSize', 12);
78     title('Consumer Choice', 'FontSize', 14);
79 end
80

```